

Operating Systems Project

Implementing a Simple HTTP Server

Project members:

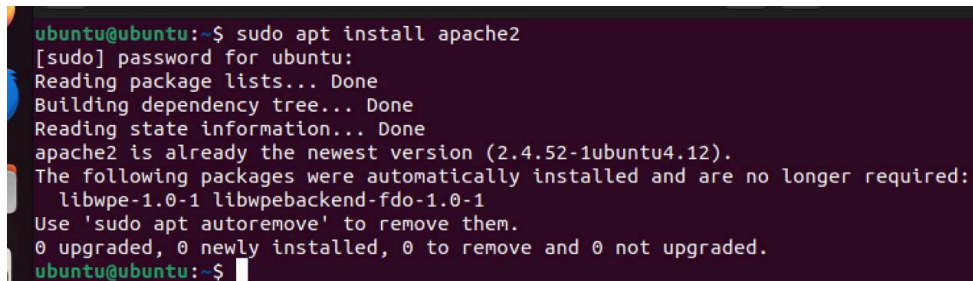
- 1.Sadeel Aldhahiri
2. Jory Alotaibi
- 3.Noura Aljasser
- 4.Layan Alharbi

Project overview:

This project involves creating a simple HTTP server using Unix command-line tools. We will install Apache, edit HTML, CSS, and JavaScript files, and serve static web content. The goal is to understand how web servers work at a basic level without complex programming.

The implementation steps:

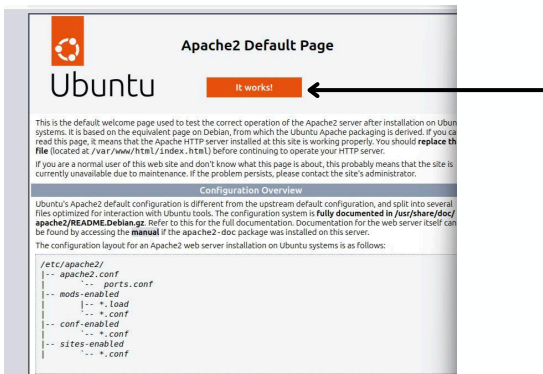
1-Install Apache



```
ubuntu@ubuntu:~$ sudo apt install apache2
[sudo] password for ubuntu:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
apache2 is already the newest version (2.4.52-1ubuntu4.12).
The following packages were automatically installed and are no longer required:
  libwpe-1.0-1 libwpebackend-fdo-1.0-1
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
ubuntu@ubuntu:~$
```

`sudo apt install apache2` : is used to install the Apache HTTP Server on a system running Ubuntu
to enable your system to host and serve web content, allowing users to access websites and web applications

2-Access Localhost

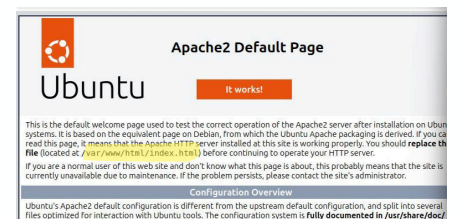


After installing apache we searched "http://localhost" in our browser if it shows this default page that says "it works", then apache installed correctly

3-Apache Default Document Root

The path /var/www/html/index.html is the default location for web files served by the Apache web server in Ubuntu, and you can change the default webpage visitors see by editing or replacing this file.

Note: The location of this path has been highlighted in yellow in the image.



4-Install Vim

```
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
ubuntu@ubuntu:~$ sudo apt install vim
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
vim is already the newest version (2:8.2.3995-1ubuntu2.19).
The following packages were automatically installed and are no longer required:
  libwpe-1.0-1 libwpebackend-fdo-1.0-1
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
ubuntu@ubuntu:~$
```

sudo apt install vim : installs the Vim text editor on an Ubuntu system to create and edit text files directly from the command line

5-Edit the HTML File

```
ubuntu@ubuntu:~$ sudo vim /var/www/html/index.html
ubuntu@ubuntu:~$
```

sudo vim /var/www/html/index.html opens the website's main HTML file in Vim with administrator permissions for editing. allows you to edit the homepage of your website directly.

```
!DOCTYPE html
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Simple HTTP Server Project</title>
  <link rel="stylesheet" href="styles.css">
  <style>
    .flowchart {
      display: flex;
      flex-direction: column;
      align-items: center;
    }
    .command {
      background-color: #005f73;
      color: white;
      border-radius: 10px;
      padding: 10px;
      margin: 10px 0;
      max-width: 90%;
      text-align: center;
    }
    .description {
      background-color: white;
      border-radius: 10px;
      padding: 10px;
      margin: 10px 0;
      max-width: 90%;
      text-align: center;
      box-shadow: 0 4px 15px rgba(0, 0, 0, 0.1);
      border-left: 5px solid #94d2bd;
    }
    .arrow {
      width: 0;
      height: 0;
      border-left: 10px solid transparent;
      border-right: 10px solid transparent;
      border-top: 10px solid #005f73;
      margin: 5px 0;
    }
    @media (max-width: 600px) {
      .command, .description {
        max-width: 100%;
      }
    }
  </style>
</head>
<body>
```

```
</style>
</head>
<body>
  <header>
    <h1>Simple HTTP Server Project</h1>
    <nav>
      <ul>
        <li><a href="#about">About</a></li>
        <li><a href="#team">Team</a></li>
        <li><a href="#commands">Commands</a></li>
        <li><a href="#supervisor">Supervisor</a></li>
        <li><a href="#contact">Contact</a></li>
      </ul>
    </nav>
  </header>
  <main>
    <section id="about">
      <h2>About</h2>
      <p>This project aims to implement a simple HTTP server.</p>
    </section>
    <section id="team">
      <h2>Our Team</h2>
      <ul>
        <li><span>Member 1: Jory Alotaibi 4441907</span></li>
        <li><span>Member 2: Layan Alharbi 444001325</span></li>
        <li><span>Member 3: Nourh Aljasser 444005293</span></li>
        <li><span>Member 4: Sadeel Aldhahiri 444002016</span></li>
      </ul>
    </section>
    <section id="commands">
      <h2>The Commands We Used</h2>
      <div class="flowchart">
        <div class="command"><code>sudo apt install apache2</code></div>
        <div class="description">Used to install the Apache HTTP Server on a system running Ubuntu</div>
        <div class="arrow"></div>
        <div class="command"><code>/var/www/html/index.html</code></div>
        <div class="description">Default location for web files served by the Apache web server in Ubuntu</div>
        <div class="arrow"></div>
        <div class="command"><code>sudo apt install vim</code></div>
        <div class="description">Installs the Vim text editor on an Ubuntu system</div>
        <div class="arrow"></div>
        <div class="command"><code>sudo vim /var/www/html/index.html</code></div>
        <div class="description">Opens the website main HTML file in vim with administrator permissions for editing. Allows you to edit the homepage of website directly</div>
        <div class="arrow"></div>
      </div>
    </section>
  </main>
</body>
```

```
<div class="command">sudo vim /var/www/html/styles.css</div>
<div class="description">Opens the styles.css file in vim with administrator permissions, used for styling your website</div>
<div class="arrow"></div>
<div class="command">sudo vim /var/www/html/script.js</div>
<div class="description">Opens javascript file for editing with administrator permissions</div>
</div>
</section>
<section id="supervisor">
  <h2>Supervisor</h2>
  <h3>Dr. Hind Alsharif</h3>
</section>
</main>
<footer>
  <p><small>© 2024 Simple HTTP Server Project</small></p>
  <script src="script.js"></script>
</body>
</html>
```

The HTML document for our “Simple HTTP Server Project” includes the following key sections:

- Project Title: The title is “Simple HTTP Server Project,” displayed at the top of the page.
- Team Members: Lists our names and student IDs.
- Commands Section: Describes the commands used in the project with explanations.
- Flowchart Structure: Organizes commands in a flowchart style for easy understanding.
- Responsive Design: The page is styled to look good on both desktop and mobile screens.
- Supervisor: Includes the name of our project supervisor.

This layout ensures the project information is clear, professional, and easy to follow.



← After editing the HTML file

6-Edit the CSS File

```
ubuntu@ubuntu:~$ sudo vim /var/www/html/styles.css
ubuntu@ubuntu:~$
```

`sudo vim /var/www/html/styles.css` : opens the styles.css file in Vim with administrator permissions, allowing you to edit the file that controls the styling of your website.

```
body {
  font-family: Arial, sans-serif;
  margin: 0;
  padding: 0;
  background-color: #f0f4f8; /* Soft light grayish blue background */
}

header {
  background: #005f73; /* Teal */
  color: white;
  padding: 20px;
  text-align: center;
  border-bottom: 5px solid #94d2bd; /* Light mint green */
}

nav a {
  color: #ee6f57; /* Soft coral */
  text-decoration: none;
  transition: color 0.3s;
}

nav a:hover {
  color: #fcbf49; /* Golden yellow on hover */
}

section {
  margin: 20px 0;
  background: white;
  padding: 15px;
  border-radius: 10px;
  box-shadow: 0 4px 15px rgba(0, 0, 0, 0.1);
  border-left: 5px solid #94d2bd; /* Light mint green */
}

h2 {
  color: #0a9396; /* Dark teal for headings */
}

footer {
  text-align: center;
  padding: 10px;
  background: #001219; /* Very dark blue */
  color: white;
  position: relative;
  bottom: 0;
  width: 100%;
}
```

CSS code defines the styling for web page:

1. Chooses the font and background for the page.
2. Designs the top part with specific colors.
3. Makes links easy to use.
4. Shapes the content areas to look organized.
5. Gives a special color to smaller titles.
6. Designs the bottom of the page.

7-Edit the JavaScript File

```
ubuntu@ubuntu:~$ sudo vim /var/www/html/script.js
ubuntu@ubuntu:~$
```

```
document.addEventListener('DOMContentLoaded', () => {
  console.log('Simple HTTP Server Project Loaded');
});
```

The command `sudo vim /var/www/html/script.js` is used to open and edit a JavaScript file located in the system's web directory with administrative permissions. By running this command, you can edit the script.js file with full administrator rights, allowing you to modify the website's JavaScript code and save the changes, even in a directory that would normally be restricted.

Final Result:

← → ↻

localhost

☆

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Simple HTTP Server Project

About

This project aims to implement a simple HTTP server.

Our Team

- Member 1: Jory Alotaibi 44411907
- Member 2: Layan Alharbi 444001325
- Member 3: Noura Aljasser 444005293
- Member 4: Sadeel Aldahiri 444002616

The Commands We Used

sudo apt install apache2

Used to install the Apache HTTP Server on a system running Ubuntu

▼

/var/www/html/index.html

Default location for web files served by the Apache web server in Ubuntu

▼

sudo apt install vim

Installs the Vim text editor on an Ubuntu system

▼

sudo vim /var/www/html/index.html

Opens the website main HTML file in vim with administrator permissions for editing. Allows you to edit the homepage of website directly

▼

sudo vim /var/www/html/styles.css

Opens the styles.css file in vim with administrator permissions, used for styling your website

▼

sudo vim /var/www/html/script.js

Opens javascript file for editing with administrator permissions

Supervisor

Dr. Hind Alsharif

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Supervisor
Dr. Hind Al-Sharif